

RAHUL RAMESH OMALUR

Atlanta, Georgia . US Citizen

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Education

Indraprastha Institute of Information Technology (IIITD)

Nov. 2022 – Present

Bachelor of Technology in Computer Science and Artificial Intelligence

New Delhi, Delhi

CGPA: 8.69

Relevant Coursework

- | | | |
|-------------------------------|------------------------|-------------------------------|
| • Large Language Models | • Machine Learning | • Computer Vision |
| • Deep Learning | • Software Development | • Operating Systems |
| • Natural Language Processing | • MLOps | • Database Management Systems |

Experience

CodePlex Technologies Private Ltd.

May 2024 – July 2024

Software Engineer Intern

Bangalore, Karnataka

- Built a **PHP Laravel** web application to automate scraping and processing of user-submitted obituary data, integrating **2–3 vendor-specific source websites daily** into a unified pipeline.
- Engineered **vendor-specific** web scrapers using **Simple HTML DOM Parser**, handling site-by-site structural variation to migrate legacy datasets into structured **MySQL** repositories.
- Diagnosed and resolved data pipeline anomalies to maintain schema consistency and accuracy across continuously ingested cross-platform records.

Projects

SmartReach AI – Email Campaign Manager | *Django, Next.js, LLMs, scikit-learn*

Jan 2025 – May 2025

- Built an end-to-end email marketing platform with **LLM-powered content generation, regression-based send-time optimization**, and a real-time analytics dashboard across the full campaign lifecycle.
- Architected a decoupled **Django REST** backend and **Next.js** frontend with JWT authentication, contact list segmentation, and campaign scheduling, supporting the full create-send-track workflow.
- Integrated compliance with **India's DPDP Act (2023)** for data handling and built unsubscribe/consent flows, making the platform production-ready for regulated marketing use cases.

SentinelAI – Intrusion Detection System | *Flask, Redis, MongoDB, OpenCV, PyTorch, Next.js*

Jan 2026 – May 2026

- Architected a **Flask REST API** backend with **Redis** face-embedding cache and **MongoDB** for persistent identity storage, serving a real-time threat-scoring pipeline across a **Next.js** dashboard.
- Engineered an end-to-end CV inference pipeline (face recognition, ByteTrack person tracking, action recognition) with **ONNX Runtime** quantization on Raspberry Pi 4, cutting memory by **40–55%** and sustaining real-time throughput.
- Reduced known-subject API response latency from **51.7ms** to **23.9ms** via Redis caching, achieving **85.06% identity accuracy** across the full system.

Nexus Agent – AI Network Simulation Analyst | *LangGraph, Next.js, Flask, SQLite, MCP*

Jan 2026 – May 2026

- Developed a **Flask REST API** backend with **workspace-isolated SQLite** instances and a **Next.js** frontend, enabling multi-tenant natural language querying over AstraSim network simulation logs.
- Implemented schema-aware **Text-to-SQL** with execution-guided self-correction via **Groq API** (sub-second inference), achieving **90–100% SQL accuracy** across query complexity levels on a **1,000-run** simulation dataset.
- Designed a **LangGraph** multi-agent system (Supervisor + SQL/Viz/Insight specialists) with sandboxed Matplotlib plot generation and graceful Fast/Thinking mode degradation for reliability under load.

Technical Skills

Languages: Python, SQL, Java, C, PHP

Frameworks & Libraries: Hugging Face, FastAPI, Flask, Django, Laravel, PostgreSQL, MongoDB, PyTorch, TensorFlow, LangChain, MCP, DSPy

Cloud & Tools: AWS, GCP, Groq, Git, Docker, Kafka, Spark, n8n, Redis, Celery

Leadership / Extracurricular

AudioBytes

Nov 2023 – May 2026

Co-ordinator

- Organized musical and technical events for college fests, coordinating logistics across teams.
- Led the college band for various inter-college competitions and festivals.